Tutorial - 8 - 24/03/26

(Decision Tree)

Email spam Detection using Decision Tree

Training Dataset

1Q. which attribute should be selected as the root node? Justify your answer with calculations.

2Q. Build the complete Decision Tree using the ID3 algorithm (use Information Gain). Draw the tree structure with all nodes, branches and leaf nodes clearly labeled.

3Q. What is the depth of your decision tree? How many leaf nodes does it have?

ID3 -> Entropy & Information Gain

(IG) -> Information Gain of each feature.

IG -> max -> best feature.

Y-class

$$\log_2 P_i = \frac{\log_{10} P_i}{\log_{10} 2} = \frac{\ln P_i}{\ln 2}$$

Entropy -> $-\sum_{i=1}^{N} P_i * \log_2(P_i)$

IG = Entropy (parent)

$$- \sum_{k} \frac{N_k}{N} \text{ Entropy}(k)$$

① ⇒ Entrophy (parent or Y)

$$= -\sum_{i=1}^{N} P_i \times \log_2(P_i)$$

$P_{spam} = \frac{5}{10} = 0.5$

$P_{Nspam} = \frac{5}{10} = 0.5$

$$= -[(0.5)\log_2(0.5) + (0.5)\log_2(0.5)]$$

$$= 1$$

⇒ Entropy (money keywords) = 0

⇓

calculating based on features

$$\text{Entropy} = -\sum_{k} \frac{N^k}{N} \text{Entropy}(k)$$

features

features

$m_k = 1 \rightarrow$ no spam
$\rightarrow$ spam

$m_k = 0 \rightarrow$ not spam
$\rightarrow$ no spam & nature of output so it is perfect classifier.

⇒ ~~IG(M/K) = 1 + 0/4~~

⇒ Entropy (money keywords) = 0

IG (m.k) = 1 − 0 = 1

2) Entropy (sender known) =

1 → spam (1)
→ Non spam (5)

0 → spam (4)
→ Non spam (0)

$$\Rightarrow -\left[\frac{1}{6}\log_2\left(\frac{1}{6}\right) + \frac{5}{6}\log_2\left(\frac{5}{6}\right) + 0 + \underbrace{\frac{4}{4}\log_2\left(\frac{4}{4}\right)}_{0}\right]$$

$$= 0.65 + 0$$

$$= 0.65$$

$IG = 1 - \left(\frac{6}{10} \times 0.65 + \frac{4}{10} \times 0\right)$

$= 0.61$

Entropy (Has Attachments)

spam = 2
Non spam = 3 } Entropy > 0

0

spam = 3
Non - spam = 2 } Entropy > 0

1

so both entropies are > 0 so that IG is < 1

1) Entropy (suspicious word)
as it has random values so Entropy > 0 and IG < 1

<u>Now</u>

Money keywords - (Root Node)

IF = 1 → spam

IF = 0 → nonspam

Depth = 1 [it is only dividing on (1/0)]

Leaf Nodes = 2 [output]